# PHENOBARBITAL

Respiratory depression
Controlled drug

**CHEMICAL NAME** 5-ethyl-5-phenylpyrimidine-2,4,6(1H,3H,5H)-trione (phenobarbitone)
**CLASS** Barbiturate antiepileptic
**TRADE NAMES** Luminal · Gardenal · Phenobarbitone sodium injection

## THERAPEUTIC USES / INDICATIONS

- Neonatal seizures (first-line).
- Status epilepticus (second/third-line).
- All seizure types except absence (low-resource settings).
- Alcohol/benzodiazepine withdrawal (some protocols).

## ADMINISTRATION

- **IV** slowly (diluted 1:10 with water for injection) max 1 mg/kg/min (≤100 mg/min) with airway support.
- **PO** at bedtime (sedation).
- **IM** deep (avoid intra-arterial).

## DOSAGES

- **Status epilepticus:** 10–20 mg/kg IV (max 1 g).
- **Neonatal seizures:** 20 mg/kg IV loading, then 2.5–5 mg/kg/day.
- **Maintenance (adult):** 60–180 mg at night.
- **Target level:** 15–40 mg/L.

## PHARMACOKINETICS

- **Onset:** IV 5–20 min; PO 1 h. **Duration:** 10–12 h+.
- Metabolised in liver (CYP2C9, 2C19); potent enzyme inducer.
- Excreted in urine (↑ with alkaline urine). **t½:** 75–120 h (adults); longer in neonates.

## PHARMACODYNAMICS

- Prolongs opening of **GABA-A chloride channels** (and directly opens at high doses); ↓ glutamate effects → anticonvulsant, sedative.

## SIDE & ADVERSE EFFECTS

### SIDE EFFECTS (COMMON)

- Sedation, drowsiness.
- Hyperactivity, irritability (children); cognitive impairment.
- Ataxia, nystagmus.

### ADVERSE EFFECTS (SERIOUS)

- Respiratory depression, hypotension (IV).
- Dependence; withdrawal seizures.
- SJS; megaloblastic anaemia, osteomalacia.
- Teratogenic (cardiac malformations); neonatal withdrawal.

## CONTRAINDICATIONS & CAUTIONS

- Acute porphyria.
- Severe respiratory depression; severe hepatic impairment.
- **Caution:** elderly, depression, pregnancy.

### KEY INTERACTIONS

- Enzyme inducer: ↓ oral contraceptives, warfarin, steroids, many drugs.
- **Alcohol, opioids, benzodiazepines:** additive CNS/respiratory depression.
- **Valproate:** ↑ phenobarbital levels.

## NURSING MANAGEMENT

- IV: monitor RR, $SpO_2$, BP; airway equipment ready.
- Monitor levels, sedation, seizure frequency.
- Neonates: monitor breathing and feeding.
- Taper slowly when stopping.

## PATIENT EDUCATION

- Causes drowsiness; no driving or alcohol.
- Never stop suddenly.
- Hormonal contraception unreliable.
- Store safely; controlled medicine.

**KEY POINTS**

- Barbiturate — GABA-A potentiator.
- Respiratory depression, sedation.
- Neonatal seizures first-line; status 20 mg/kg IV.
- Very long t½; enzyme inducer.